Derivatives Sketchbook

10 classes · Derivatives · Fall 2026 finals

Each class: the idea in one line, the sketch step by step, plain explanation, formulas, a worked example, common traps and where to read more.

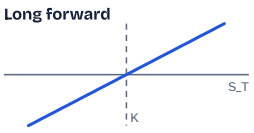
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The four derivatives and their payoffs

A derivative is a contract whose value comes from something else (a stock, a rate, oil). There are four basic kinds: forwards, futures, swaps and options.

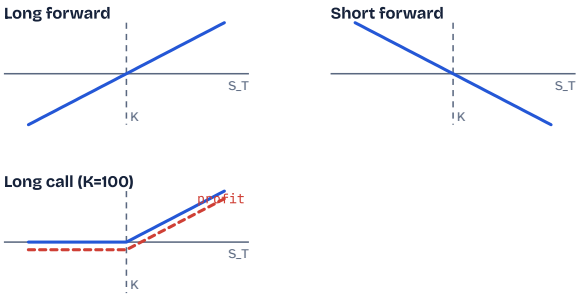
The sketch, step by step



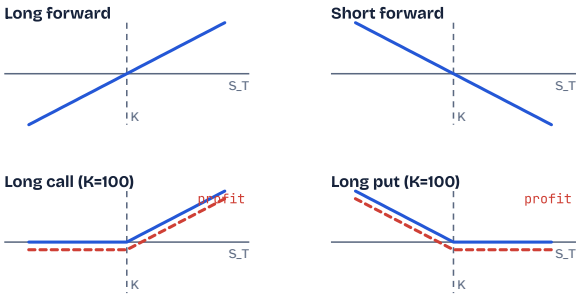
Step 1. Long forward: you win $1 for every $1 the price ends above K, and lose the same below.



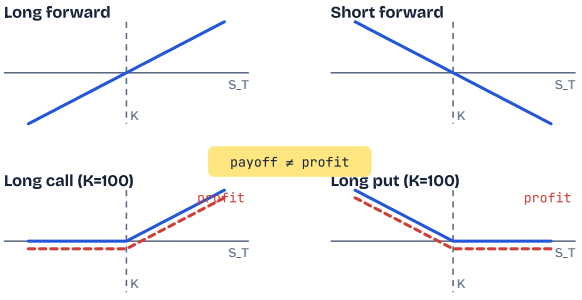
Step 2. Short forward: the mirror image. Your gain is exactly the other side's loss.



Step 3. Long call: you only use it if the price is above K. Dashed line = profit after paying the premium.



Step 4. Long put: you profit when the price falls below K. Insurance for a stock owner.



Step 5. Four shapes to remember: two straight lines, two "hockey sticks".

What it means

- **Forward:** agree today to buy (long) or sell (short) at a fixed price K on a later date. Both sides must go through with it. Made privately (over the counter).
- **Future:** the same idea, but traded on an exchange, standardized, and settled every day through margin (Class 7).
- **Swap:** trade one stream of payments for another, e.g. fixed interest for floating (Class 9).
- **Option:** the *right but not the obligation*. A call is the right to buy at K, a put the right to sell at K. You pay a premium up front for that right.

Payoff is what you get at the end. **Profit** is payoff minus what you paid. Draw the picture every time; it stops sign mistakes.

The formulas

```
Long forward payoff  = S_T - K
Short forward payoff = K - S_T
Long call payoff     = max(S_T - K, 0)
Long put payoff      = max(K - S_T, 0)
Profit = payoff - premium paid
```

S_T = price of the asset at the end, K = agreed (strike/delivery) price.

Worked example

Question. You buy a call with strike $100 for a $5 premium. At expiry the stock is $120. What is your payoff and profit? What if the stock is $90?

1. At $120: payoff = max(120 − 100, 0) = `20`
2. Profit = 20 − 5 = `15`
3. At $90: payoff = max(90 − 100, 0) = `0` (you just do not use the option)
4. Profit = 0 − 5 = `-5`. The most a call buyer can lose is the premium.

Answer: At $120: payoff 20, profit 15. At $90: payoff 0, profit −5

Common traps

- Mixing payoff with profit.
- Saying a forward costs money to enter. A fair forward costs nothing up front.
- Forgetting the short side of an option can lose a lot.

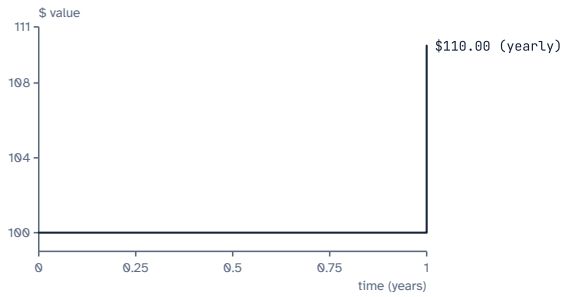
Where to read more

- `Derivatives/03 Lectures/FRE6073_LECTURE_02_Lecture_01_Introduction_to_Derivatives.pdf`
- `Hull 8e, Chapters 1 and 10`

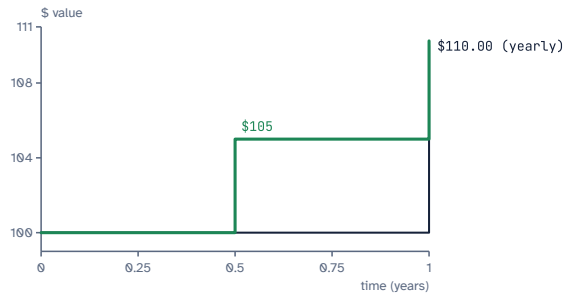
How often interest is added

"10% a year" can mean different amounts of money depending on how often interest is added. Continuous compounding is the limit where it is added every instant.

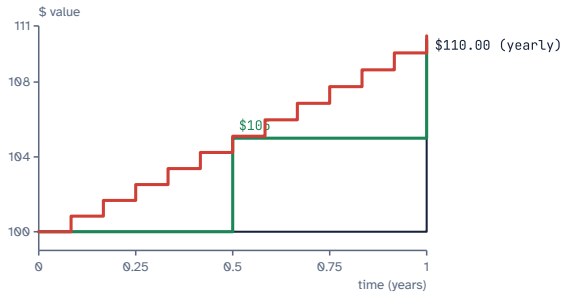
The sketch, step by step



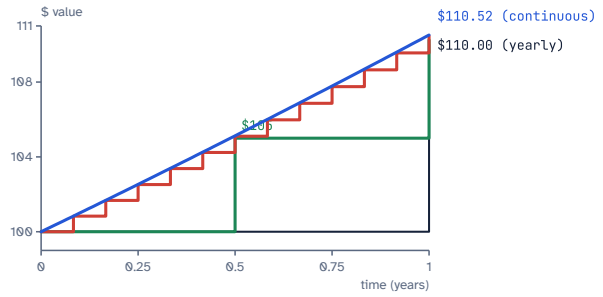
Step 1. $100 at 10% a year. Added once a year, it jumps to $110 at the end.



Step 2. Added twice a year: $105 after 6 months, then 5% on $105 gives $110.25.



Step 3. Monthly: 12 tiny jumps. You end at $110.47.



Step 4. Continuous: a smooth curve, interest every instant. You end at $110.52.



Step 5. The more often you compound, the more you get, but it levels off at e^0.10.

What it means

If the bank adds interest twice a year, you earn interest on your interest sooner, so you end up with a bit more. Same headline rate, different result. That is why every rate needs a **compounding frequency** attached.

- **m times a year:** grow by (1 + R/m) each piece, m pieces a year.
- **Continuous:** grow by e^{Rt}. Derivatives math (forwards, Black-Scholes) almost always uses this.
- **Converting:** two rates are "the same" if they grow $1 to the same amount in one year.

The formulas

```
Compounded m times:  FV = A × (1 + R/m)^(m·t)
Continuous:          FV = A × e^(R·t)
Convert m → cont:    Rc = m × ln(1 + Rm/m)
Convert cont → m:    Rm = m × (e^(Rc/m) - 1)
```

A = amount, R = annual rate, m = times per year, t = years, ln = natural log.

Worked example

Question. You have 10% a year compounded twice a year. What is the continuously compounded rate?

1. Each half year you grow by 1 + 0.10/2 = `1.05`
2. Use the convert rule: `Rc = 2 × ln(1.05)`
3. ln(1.05) = 0.04879, times 2 = `0.09758`
4. Check: e^0.09758 = 1.1025, and 1.05^2 = 1.1025. Same growth.

Answer: Rc = 9.758% continuous

Calculator example

Inputs: Annual rate (%) = 10, Times per year = 2

```
$100 after 1 year = 100 × (1 + 0.1/2)^2 = $110.25
Same as continuous rate Rc = 2 × ln(1 + 0.0500) = 9.758%
```

Common traps

- Plugging a semiannual rate into e^(rT) without converting it.
- Using t in months when the rate is per year.
- Thinking a higher m changes the headline rate. It only changes how much you actually earn.

Where to read more

- `Derivatives/03 Lectures/FRE6073_LECTURE_05_Lecture_04_Interest_Rates.pptx`
- `Hull 8e, Chapter 4 (Interest Rates)`

Forward price by no-arbitrage

The fair forward price is what it costs to buy the asset now with borrowed money and carry it to the delivery date. Any other price gives someone free money.

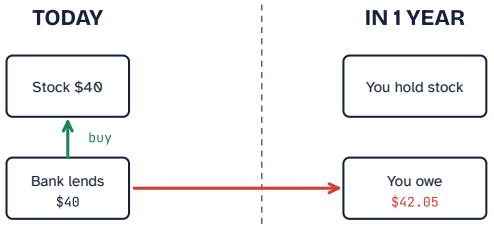
The sketch, step by step



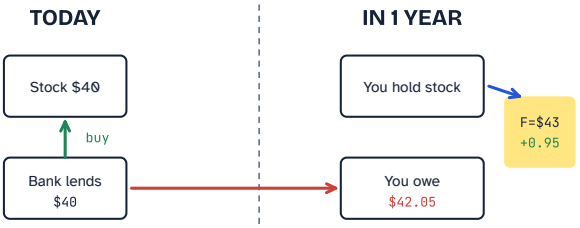
Step 1. Today the stock costs $40. You can borrow at 5%.



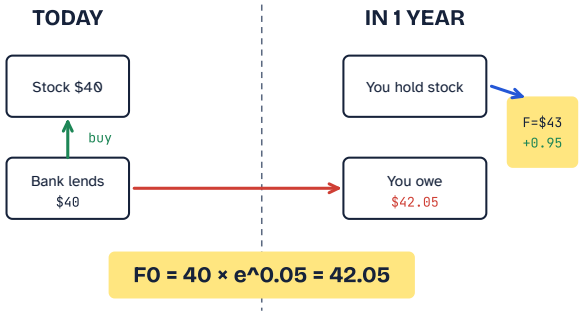
Step 2. Strategy: borrow $40 and buy the stock. Costs you nothing today.



Step 3. In one year you own the stock and owe 40 × e^0.05 = $42.05.



Step 4. If someone offers a forward at $43, sell it to them. Deliver the stock, get 43, repay 42.05. Free $0.95.



Step 5. So the only price with no free money is F0 = $42.05.

What it means

Arbitrage means a riskless profit with no money down. Markets hunt it away. So: if you can borrow $40 at 5%, buy the stock, and hold it a year, it costs you $40 × e^0.05 = $42.05 to own the stock in a year. A forward to buy it in a year must also cost $42.05, otherwise you would do one and the opposite of the other.

- **Income** (dividends) makes holding the stock cheaper, so subtract its PV: $F = (S - I)e^{rT}$.
- **Dividend yield q** or a foreign interest rate r_f works the same: $F = S\,e^{(r-q)T}$.

- **Forward price vs forward value:** a new forward is worth 0. An old one with delivery price K is worth $(F_0 - K)e^{-rT}$ to the long side.

The formulas

```
No income:        F0 = S0 × e^(rT)
Known income I:   F0 = (S0 - I) × e^(rT)
Yield q / FX:     F0 = S0 × e^((r - q)T)     (q = r_f for currency)
Value of long:    f  = (F0 - K) × e^(-rT)
```

S0 = spot price now, r = risk-free rate (continuous), T = years, K = delivery price written in an existing contract.

Worked example

Question. A stock is $40, pays no dividend, r = 5%, T = 1 year. (a) Fair forward price? (b) The market quotes $43. How do you make money? (c) An old contract has K = $40. What is it worth?

1. (a) F0 = 40 × e^0.05 = `42.05`
2. (b) $43 is too high. Sell the forward at 43, borrow $40, buy the stock.
3. In a year: deliver the stock, receive 43, repay the loan 42.05. Profit `0.95` with no risk.
4. (c) f = (42.05 − 40) × e^(−0.05) = `1.95` for the long side.

Answer: F0 = 42.05; arbitrage profit 0.95; old contract worth 1.95

Calculator example

Inputs: Spot price S0 = 40, Rate r (%) = 5, Yield q or r_f (%) = 0, Years T = 1, Old contract K = 40

```
Fair forward F0 = 40 × e^(0.0500 × 1) = 42.0508
Value of old long contract = (42.0508 - 40) × e^(-0.05 × 1) = 1.9508
```

Common traps

- Mixing price (the K that makes value 0) and value (what an existing contract is worth).
- Adding dividends instead of subtracting them.
- Doing the arbitrage in the wrong direction. Forward too high means sell the forward and buy the asset.

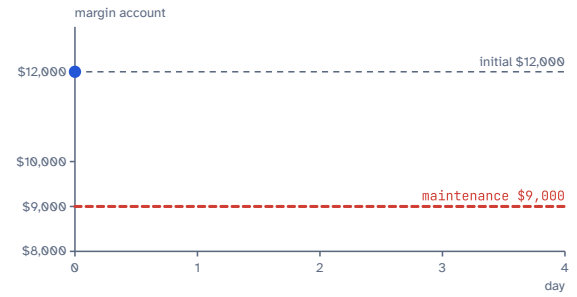
Where to read more

- `Derivatives/03 Lectures/FRE6073_LECTURE_03_Lecture_02_Forward_and_Futures_Prices.pptx`
- `Derivatives/04 Assignments and Solutions/FRE6073_SOLUTION_03_Assignment_01_Solutions.pdf`

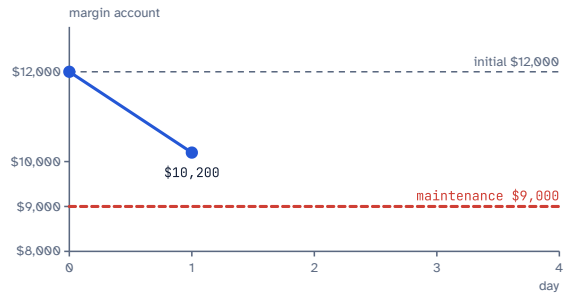
Futures, margin and daily settlement

With futures, gains and losses are paid every day through a margin account, so nobody can build up a big unpaid loss.

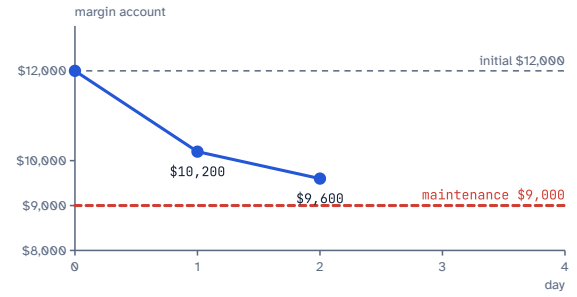
The sketch, step by step



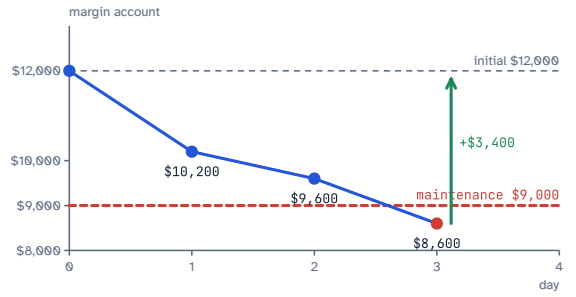
Step 1. You open with $12,000. The dashed red line is the $9,000 maintenance floor.



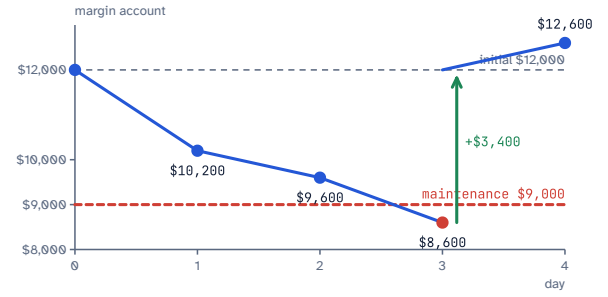
Step 2. Day 1: gold falls $9. You lose $1,800 that day. Balance $10,200.



Step 3. Day 2: down $3 more. Balance $9,600.



Step 4. Day 3: down $5. Balance $8,600 is under the floor. Margin call! Add $3,400 to get back to $12,000.



Step 5. Day 4: gold rises $3. You gain $600. Balance $12,600.

What it means

- **Initial margin:** the deposit to open the position.
- **Marking to market:** each day the exchange moves money between buyers and sellers based on the new futures price.
- **Maintenance margin:** a floor. If your account drops below it you get a **margin call** and must top it back up to the *initial* margin.
- **Clearing house:** stands between buyer and seller, so you never depend on the other trader paying. That removes most counterparty risk.
- **Basis** = spot price − futures price. It goes to zero at delivery.

The formulas

```
Daily gain (long) = (F_today - F_yesterday) × contract size × number of contracts
If balance < maintenance → deposit (initial - balance)
```

Short positions get the opposite sign.

Worked example

Question. You go long 2 gold futures (100 oz each) at $1,250. Initial margin $6,000 per contract, maintenance $4,500. Prices: 1,241 → 1,238 → 1,233 → 1,236. Track the account.

1. Start: 2 × 6,000 = `$12,000`. Each $1 move = 2 × 100 = $200.
2. Day 1: −9 × 200 = −1,800 → `$10,200`
3. Day 2: −3 × 200 = −600 → `$9,600`
4. Day 3: −5 × 200 = −1,000 → `$8,600`, below the $9,000 floor. Margin call: deposit `$3,400` to get back to $12,000.
5. Day 4: +3 × 200 = +600 → `$12,600`

Answer: Margin call of $3,400 on day 3; balance $12,600 after day 4

Common traps

- Topping up only to the maintenance level. You go back to the initial margin.
- Forgetting to multiply by contract size and number of contracts.
- Thinking margin is a fee. It is your money, held as a deposit.

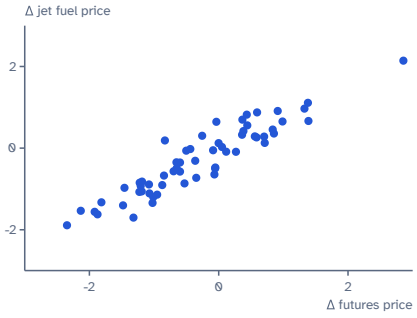
Where to read more

- `Derivatives/03 Lectures/FRE6073_LECTURE_04_Lecture_03_Futures_Markets_and_Margins.pptx`
- `Hull 8e, Chapter 2`

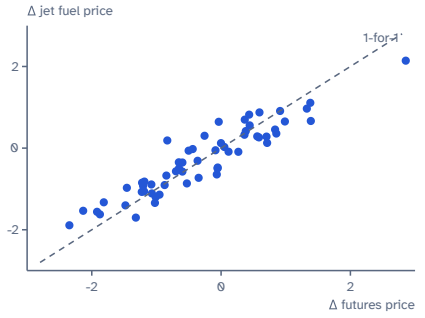
Hedging with futures

A hedge uses futures to cancel out a price risk you already have. The hedge ratio tells you how many futures to use when the futures and your exposure do not move one-for-one.

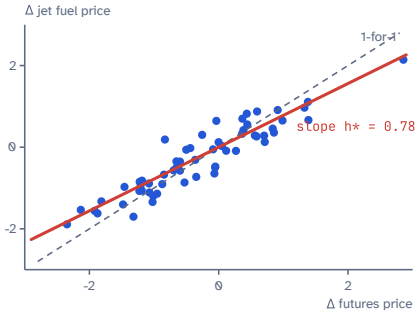
The sketch, step by step



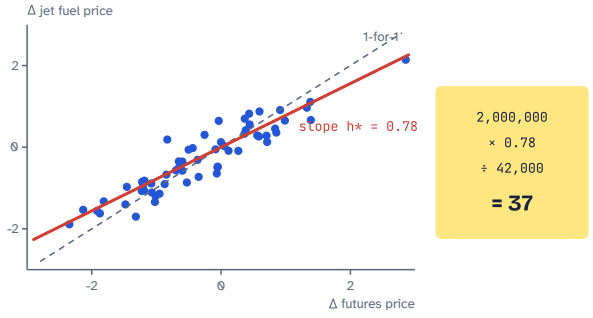
Step 1. Each dot is one month: how much jet fuel moved (up) vs how much heating oil futures moved (across).



Step 2. They move together, but not exactly one-for-one.



Step 3. The best-fit line has slope 0.78. That slope is the hedge ratio h*.



Step 4. So hedge 78% of the 2m gallons: 1.56m gallons ÷ 42,000 = 37 contracts.

What it means

- **Short hedge:** you will sell something later (a farmer's wheat). Sell futures now so a price fall is offset by a futures gain.
- **Long hedge:** you will buy something later (an airline's fuel). Buy futures now.
- **Basis risk:** basis = spot − futures. If you hedge with a different but related product (heating oil for jet fuel), or close out early, the basis can move and the hedge is not perfect.
- **Minimum-variance hedge ratio h*:** the slope when you regress changes in your price on changes in the futures price. It is $\rho \times \sigma_S/\sigma_F$.
- **Number of contracts:** h* × (size of your exposure) / (size of one contract).
- **Stock index hedge:** to remove market risk from a portfolio with beta β, short β × (portfolio value)/(futures price × multiplier) contracts.

The formulas

```
h* = ρ × σS / σF
N* = h* × QA / QF
Index hedge: N* = β × VA / VF
```

σS, σF = std dev of price changes (spot, futures), ρ = their correlation, QA = units exposed, QF = units per contract, VA = portfolio value, VF = futures price × multiplier.

Worked example

Question. An airline will buy 2 million gallons of jet fuel. It hedges with heating oil futures (42,000 gallons each). σS = 0.0263, σF = 0.0313, ρ = 0.928. How many contracts?

1. h* = 0.928 × 0.0263 / 0.0313 = `0.7798`
2. Exposure to hedge = 0.7798 × 2,000,000 = `1,559,520 gallons`
3. Contracts = 1,559,520 / 42,000 = `37.13`
4. Round: buy `37` heating oil futures (a long hedge, since it will buy fuel).

Answer: h* ≈ 0.78, buy about 37 contracts

Calculator example

Inputs: Correlation ρ = 0.928, σ spot = 0.0263, σ futures = 0.0313, Units exposed = 2000000, Units per contract = 42000

```
h* = 0.928 × 0.0263 / 0.0313 = 0.7798
Contracts = 0.7798 × 2,000,000 / 42,000 = 37.13
```

Common traps

- Using h = 1 when the futures is on a different product.
- Going short when you need to buy later (it is a long hedge).
- Forgetting that a hedge also removes your upside.

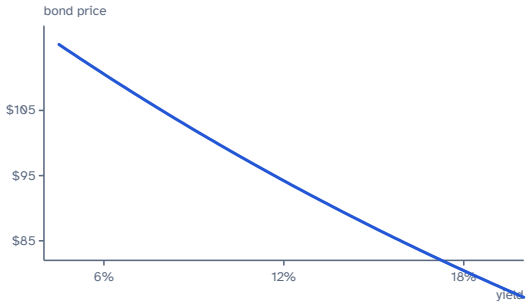
Where to read more

- `Derivatives/03 Lectures/FRE6073_LECTURE_04_Lecture_03_Futures_Markets_and_Margins.pptx`
- `Hull 8e, Chapter 3 (Hedging strategies using futures)`

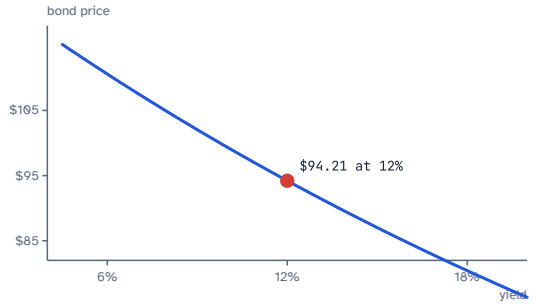
Forward rates, duration and convexity

Duration tells you how much a bond price drops when rates rise. Convexity is the correction, because the real price curve bends.

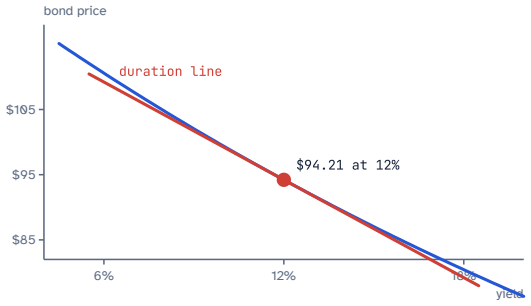
The sketch, step by step



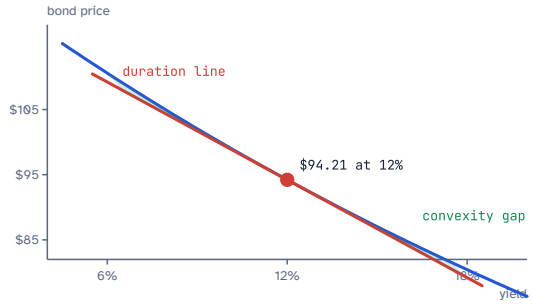
Step 1. The price of the bond at each yield. It slopes down and bends (is "convex").



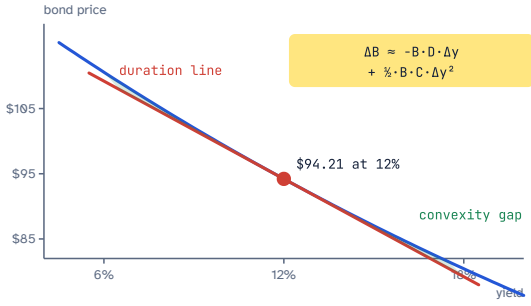
Step 2. At today's yield of 12%, the price is $94.21.



Step 3. Duration draws the tangent line at that point. Slope = −B × D.



Step 4. The tangent is below the curve on both sides. That gap is convexity.



Step 5. For a small move the line is close. For a big move, add the convexity term or reprice fully.

What it means

Forward rate: the rate for a future period that today's zero rates lock in. If 1 year pays 3% and 2 years pays 4%, the second year alone must pay 5% (continuous rates).

Duration is the weighted average time you wait for your cash, where the weights are each payment's share of the price. It is also the sensitivity: a 1% rise in yield cuts the price by about D%.

Convexity: the price-yield curve is bowed. A straight-line (duration) estimate always *underestimates* the price after a big move either way. Full repricing is the true answer; duration

is the quick estimate.

The formulas

```
Forward rate:   F(T1,T2) = (R2·T2 - R1·T1) / (T2 - T1)
Duration:       D = Σ t × [CF(t)·e^(-y·t)] / B
Price change:   ΔB ≈ -B × D × Δy  (+ ½ × B × C × Δy²)
```

R = zero rates, y = bond yield (continuous), B = bond price, C = convexity.

Worked example

Question. 3-year bond, 10% coupon paid twice a year, yield 12% continuous (Hull's example). Price, duration, and the effect of a 0.1% rise in yield?

1. Cash flows: $5 at 0.5, 1, 1.5, 2, 2.5 years and $105 at 3 years.
2. Discount each at e^(−0.12t) and add: `B = 94.213`
3. Weight each time by its share of the price and add: `D = 2.653 years`
4. ΔB ≈ −94.213 × 2.653 × 0.001 = `-0.250`, so the price falls to about 93.963.

Answer: B = 94.21, D = 2.653, price drops about $0.25

Common traps

- Using Δy = 0.1 instead of 0.001.
- Forgetting the minus sign: rates up means price down.
- Using duration for a large rate shock without the convexity term.

Where to read more

- `Derivatives/03 Lectures/FRE6073_LECTURE_06_Lecture_05_Duration_and_Swaps.pptx`
- `Risk syllabus RMN2 (duration, convexity, term structure)`

Swaps: trading payment streams

In an interest rate swap, one side pays fixed and receives floating. You value it as owning one bond and owing another.

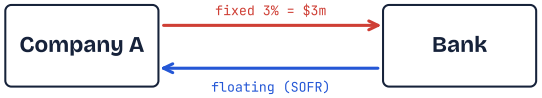
The sketch, step by step



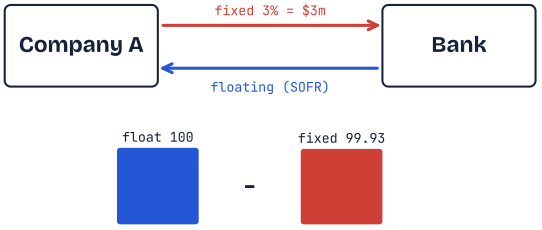
Step 1. Two parties: Company A and a bank.



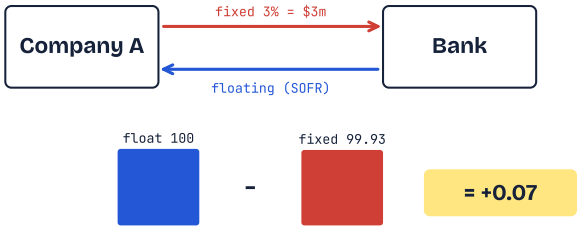
Step 2. A pays the bank a fixed 3% of $100m every year: $3m.



Step 3. The bank pays A the floating rate (e.g. SOFR) on $100m.



Step 4. Same as A owning a floating bond (worth 100) and owing a fixed bond (worth 99.93).



Step 5. Swap value to A = 100 − 99.93 = +0.07 ($m).

What it means

Company A has a floating-rate loan but wants a fixed payment. It enters a swap: pays a fixed 3% to the bank, receives the floating rate back. Only interest moves; the notional (e.g. $100m) never changes hands in an interest rate swap.

- **Bond view:** receive floating + pay fixed = long a floating bond, short a fixed bond. Value = $B_{float} - B_{fixed}$.
- Right after a reset, a floating bond is worth its par value (100).
- **Forward view:** treat each future exchange as a forward rate agreement and add them up. Both views give the same answer.

- **Currency swap:** swap interest *and* principal in two currencies. Value each leg in its own currency, convert at today's spot rate.

The formulas

```
Pay fixed, receive floating:   V = B_float - B_fixed
B_fixed = Σ c·e^(-r_i·t_i) + L·e^(-r_n·t_n)
B_float = L  (just after a reset)
```

L = notional, c = fixed coupon payment, r_i = zero rate for time t_i.

Worked example

Question. $100m swap, 2 years left, you pay 3% fixed annually and receive floating (just reset). Zero rates: 1-yr 2.5%, 2-yr 3.0% (continuous). Value to you?

1. Fixed bond: 3 × e^(−0.025) + 103 × e^(−0.06)
2. = 2.926 + 97.001 = `99.928` ($m)
3. Floating bond right after reset = `100`
4. Value = 100 − 99.928 = `+0.072` ($m). Positive: the 3% you pay is a little below the fair swap rate for this curve (about 3.04%), so the swap is slightly in your favour.

Answer: Swap worth about +$72,000 to the fixed payer

Common traps

- Getting the sign backwards. Write which side you are on first.
- Forgetting the notional at the end of the fixed bond (103, not 3).
- For currency swaps, converting with forward rates in the bond view. Use spot.

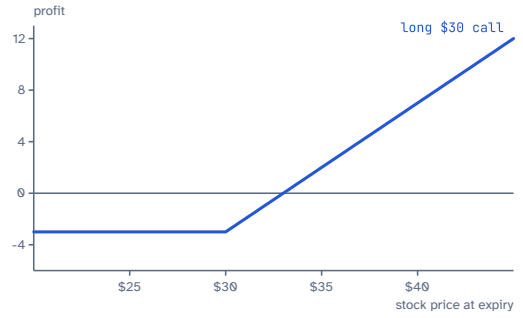
Where to read more

- `Derivatives/03 Lectures/FRE6073_LECTURE_06_Lecture_05_Duration_and_Swaps.pptx`
- `Derivatives/04 Assignments and Solutions/FRE6073_ASSIGNMENT_05_Assignment_03.pdf`

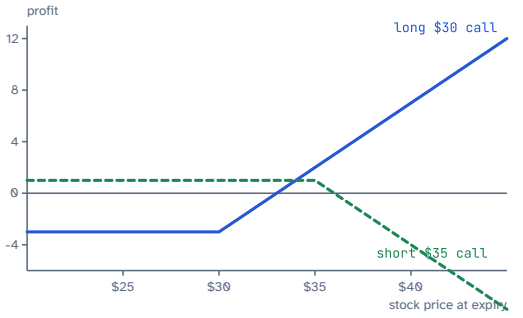
Option strategies and put-call parity

Options can be stacked like Lego to make any payoff shape you want. Put-call parity is the rule that links a call and a put with the same strike.

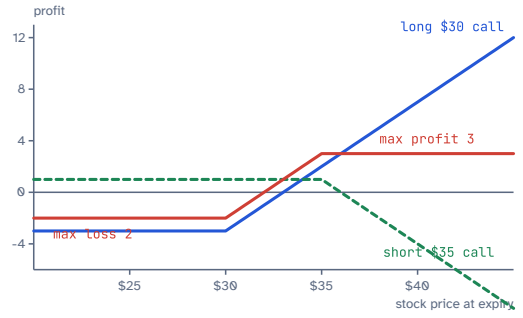
The sketch, step by step



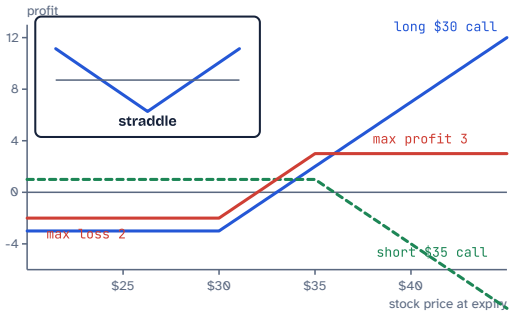
Step 1. Start: buy a $30 call. Hockey stick that starts at 30.



Step 2. Add: sell a $35 call. That flips the slope back to flat above 35.



Step 3. Together: the bull spread. Profit is capped at 3, loss is capped at 2.



Step 4. Different idea: a straddle (call + put at the same strike) is a V. It wins on big moves either way.

What it means

- **Bull call spread:** buy a low-strike call, sell a high-strike call. Cheaper than a call alone, but profit is capped. You win if the price rises.
- **Bear spread:** the opposite, for a falling price.
- **Straddle:** buy a call and a put at the same strike. You win if the price moves a lot either way. A bet on volatility.
- **Strangle:** like a straddle with two different strikes. Cheaper, needs a bigger move.
- **Butterfly:** buy calls at K1 and K3, sell two at K2. Wins if the price stays near K2 (Class "Volatility smiles" uses it to find probabilities).
- **Covered call / protective put:** stock + short call, or stock + long put (insurance).
- **Put-call parity:** a call plus cash $K \cdot e^{-rT}$ pays the same as a put plus the stock, so they must cost the same. Only for European options.
- **Bounds:** a call is worth at least $S - Ke^{-rT}$ and at most S.

The formulas

```
Put-call parity:  c + K·e^(-rT) = p + S0
Call bounds:      max(S0 - K·e^(-rT), 0) ≤ c ≤ S0
Put bounds:       max(K·e^(-rT) - S0, 0) ≤ p ≤ K·e^(-rT)
```

Profit of a strategy = sum of the payoffs − net premium paid.

Worked example

Question. (a) Buy a $30 call for $3 and sell a $35 call for $1. Max profit and max loss? (b) A European call with K = 30, T = 3 months costs $3. S = $31, r = 10%. What must the put cost?

1. (a) Net cost = 3 − 1 = 2. Max loss = 2 (price ends below 30).
2. Max payoff = 35 − 30 = 5, so max profit = 5 − 2 = 3 (price ends above 35).
3. (b) p = c + K·e^(−rT) − S = 3 + 30·e^(−0.025) − 31
4. = 3 + 29.259 − 31 = 1.26

Answer: Spread: max profit 3, max loss 2. Put = $1.26

Calculator example

Inputs: Call price c = 3, Stock S0 = 31, Strike K = 30, Rate r (%) = 10, Years T = 0.25

```
PV of strike = 30 × e^(-0.1 × 0.25) = 29.2593
Put = 3 + 29.2593 - 31 = 1.2593
Call lower bound = max(S - PV(K), 0) = 1.7407
```

Common traps

- Using put-call parity for American options (it only gives bounds there).
- Forgetting to subtract the premium when drawing profit.
- Mixing up which leg is bought and which is sold in a spread.

Where to read more

- Hull 8e, Chapters 10–11 (Properties of options, trading strategies)
- Valuation/03 Lectures/FRE6103_LECTURE_11_Options_in_a_valuation_context.pdf

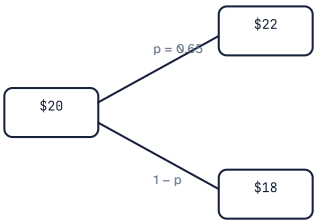
Binomial trees for options

Let the stock only go up or down each step. Then you can copy the option exactly with stock and cash, and that copy's cost is the option price.

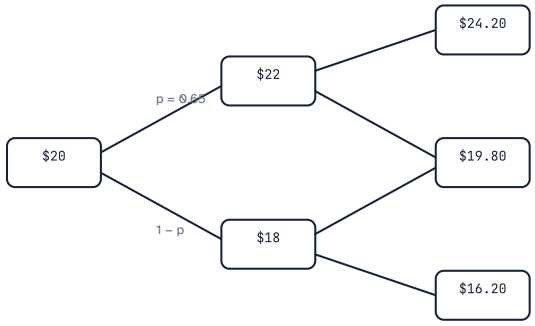
The sketch, step by step



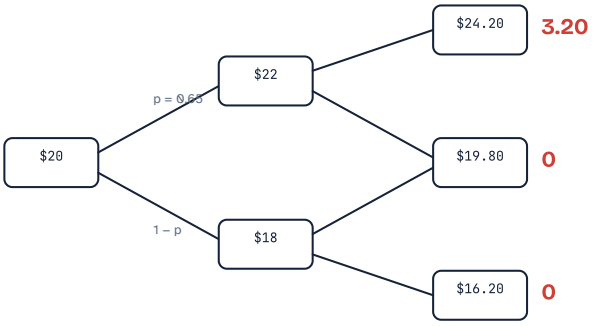
Step 1. Today the stock is $20.



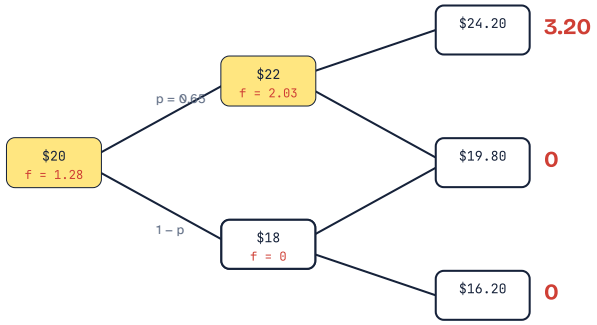
Step 2. After 3 months it is either $22 (up 10%) or $18 (down 10%).



Step 3. After 6 months: $24.20, $19.80 or $16.20.



Step 4. At the end, a call with strike $21 pays 3.20, 0, 0.



Step 5. Step back using p = 0.6523 and discount: 2.03 at the up node, then $1.28 today.

What it means

- **Replication:** hold Δ shares and borrow some cash so the portfolio pays the same as the option in both the up and down state. $\Delta = (f_u - f_d)/(S_u - S_d)$.
- **Risk-neutral probability p:** the made-up probability that makes the stock earn the risk-free rate. It is a pricing tool, not anyone's real forecast.
- **Backward induction:** start at the end with payoffs, step back one node at a time: value = $e^{-r\Delta t}[p \cdot f_u + (1-p) \cdot f_d]$.
- **American options:** at each node, value = max(hold value, exercise now). Matters most for puts.
- With many small steps the tree converges to Black-Scholes (Class 17). Typical choice: $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$.

The formulas

```
p = (e^(rΔt) - d) / (u - d)
f = e^(-rΔt) × [p·f_u + (1 - p)·f_d]
Δ = (f_u - f_d) / (S·u - S·d)
American: f = max(f, exercise value)
```

u, d = up and down factors, Δt = length of one step, f = option value at a node.

Worked example

Question. Stock $20, u = 1.1, d = 0.9, r = 12%, each step is 3 months. Price a European call with strike $21: one step, then two steps.

1. p = (e^(0.12×0.25) − 0.9) / 0.2 = (1.0305 − 0.9)/0.2 = `0.6523`
2. One step: up price 22 → payoff 1; down 18 → payoff 0. c = e^(−0.03) × 0.6523 × 1 = `0.633` (Δ = 1/4 = 0.25 shares)
3. Two steps: ends 24.2, 19.8, 16.2 → payoffs 3.2, 0, 0
4. Up node: e^(−0.03) × 0.6523 × 3.2 = `2.0257`; down node = 0
5. Today: e^(−0.03) × 0.6523 × 2.0257 = `1.2822`

Answer: One-step call = $0.633; two-step call = $1.2822

Calculator example

Inputs: Stock S = 20, Up factor u = 1.1, Down factor d = 0.9, Rate r (%) = 12, Step length (years) = 0.25, Strike K = 21

```
p = (1.0305 - 0.9) / (1.1 - 0.9) = 0.6523
Call payoffs: up 1.00, down 0.00 → call = 0.6330, Δ = 0.2500
Put payoffs: up 0.00, down 3.00 → put = 1.0124
```

Common traps

- Using the real-world probability instead of p.
- Discounting by the whole T at each step instead of Δt.
- For American puts, forgetting to compare with early exercise at every node.

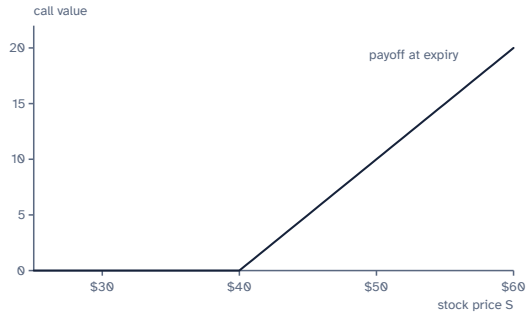
Where to read more

- `Derivatives/03 Lectures/FRE6073_LECTURE_01_Binomial_trees.pdf`
- `Hull 8e, Chapter 12`

Black-Scholes and the Greeks

Black-Scholes gives a formula for a European option price from five inputs. The Greeks say how the price moves when one input moves.

The sketch, step by step



Step 1. At expiry a call with strike $40 is worth max(S − 40, 0): a hockey stick.



Step 2. Before expiry the price is a smooth curve above the stick. The gap is time value.



Step 3. At S = $42 with 6 months left, Black-Scholes says $4.76.



Step 4. The slope of the curve there is delta = 0.78. Move the stock $1, the call moves about $0.78.



Step 5. With 3 months left (dashed) the curve sits lower and hugs the stick. That loss over time is theta. How sharply the curve bends is gamma.

What it means

Idea behind it: hold the option and short Δ shares. Over a tiny instant the randomness cancels, so the hedge must earn the risk-free rate. That gives a PDE whose answer is the formula. Prof. Shimko's "Simpler Option Pricing" lecture reaches the same formula with only college calculus.

- **Inputs:** stock S, strike K, rate r, volatility σ, time T. Note: the expected return μ is *not* an input.
- **Delta** N(d1): change in option price per $1 move in the stock. Also the hedge ratio.
- **Gamma:** how fast delta changes. High near the strike close to expiry.

- **Theta:** value lost per day as time passes. **Vega:** change per 1% more volatility. **Rho:** change per 1% higher rate.
- **Put-call parity:** $c + Ke^{-rT} = p + S$. Use it to check your answer.

The formulas

```
c = S·N(d1) - K·e^(-rT)·N(d2)
p = K·e^(-rT)·N(-d2) - S·N(-d1)
d1 = [ln(S/K) + (r + σ²/2)·T] / (σ√T)
d2 = d1 - σ√T
Delta(call) = N(d1)    Gamma = φ(d1)/(S·σ·√T)    Vega = S·φ(d1)·√T
```

N = cumulative normal (NORM.S.DIST(x,TRUE) in Excel), φ = normal density.

Worked example

Question. S = 42, K = 40, r = 10%, σ = 20%, T = 0.5 years (Hull's example). Find the call, the put and delta.

1. σ√T = 0.2 × 0.7071 = `0.1414`
2. d1 = [ln(1.05) + (0.10 + 0.02) × 0.5] / 0.1414 = (0.0488 + 0.06)/0.1414 = `0.7693`
3. d2 = 0.7693 − 0.1414 = `0.6278`; N(d1) = 0.7791, N(d2) = 0.7349
4. c = 42 × 0.7791 − 40 × e^(−0.05) × 0.7349 = `4.76`
5. Put by parity: p = 4.76 − 42 + 40e^(−0.05) = `0.81`. Delta = `0.779`

Answer: Call $4.76, put $0.81, delta 0.78

Calculator example

Inputs: Stock S = 42, Strike K = 40, Rate r (%) = 10, Volatility σ (%) = 20, Years T = 0.5

```
d1 = 0.7693, d2 = 0.6278
N(d1) = 0.7791, N(d2) = 0.7349
Call = 4.7594, Put = 0.8086
Delta = 0.7791, Gamma = 0.0500, Vega (per 1%) = 0.0881
```

Common traps

- Using σ^2 where σ goes, or forgetting √T.
- Using a non-continuous rate in e^(−rT).
- Thinking the stock's expected return changes the price. It does not.

Where to read more

- `Valuation/03 Lectures/FRE6103_LECTURE_12_Simpler_Option_Pricing_Update_2026.pdf`
- `Valuation/03 Lectures/FRE6103_LECTURE_09_Lecture_9_BSM_and_Implied_Densities.pdf`
- `Hull 8e, Chapters 14 and 18`

Watch and learn online

Free videos and courses to use alongside these classes. Coursera courses are free when you choose "audit".

Derivatives

- **Khan Academy: Finance and capital markets** (Free short videos, very beginner)
  https://www.khanacademy.org/economics-finance-domain/core-finance
  Watch the units on forwards, futures, options, put-call parity and swaps first. Best place to start if a class here feels too fast.
- **MIT OpenCourseWare 18.S096: Mathematics with Applications in Finance** (Free full lecture videos (MIT, 2013))
  https://ocw.mit.edu/courses/18-s096-topics-in-mathematics-with-applications-in-finance-fall-2013/
  Watch the lectures on stochastic processes, Itô calculus and Black-Scholes once you finish the Binomial and Black-Scholes classes here.
- **Columbia (Coursera): Introduction to Financial Engineering and Risk Management** (Free to audit, video course)
  https://www.coursera.org/learn/financial-engineering-intro
  Covers no-arbitrage, forwards, swaps, binomial trees and option pricing in the same order as your course.
- **Bionic Turtle (YouTube)** (Free short videos on Hull topics)
  https://www.youtube.com/@bionicturtle
  Search the channel for any Hull chapter title (e.g. "hedge ratio", "binomial", "greeks") for a 5–10 minute explanation.